College of Engineering - Department of Electrical Engineering

OUTCOMES-BASED EDUCATION (OBE) COURSE SYLLABUS

I. COURSE INFORMATION

Course Code:	**EE 312**	**Credit Units:**	**3.0 Units (3 hrs Lecture)**
Course Title:	Feedback and Control Systems	Semester / A.Y.:	1st Semester, A.Y. 2026-2027
Section:	BSEE 3A	Faculty / Instructor:	Engr. Juan Dela Cruz, MSEE
Passing Cutoff:	60% (3.00 Passing Grade)	Grading System:	Midterm (50%) + Final (50%)

II. COURSE OUTCOMES (CO) AND KEY PERFORMANCE INDICATORS (KPI)

CO Code	**Course Outcome Statement**	**Program Outcomes**	**Target Score**	**Target Class %**
CO1	Formulate mathematical models (transfer functions & state-space equations) of electrical, mechanical, and electromechanical systems.	PO-a, PO-e	60%	70%
CO2	Analyze time-domain transient and steady-state performance and Routh-Hurwitz stability of linear feedback control systems.	PO-a, PO-b, PO-e	60%	70%
CO3	Evaluate closed-loop stability and dynamic response using Root Locus, Bode Plots, and Nyquist Frequency-Domain techniques.	PO-b, PO-e, PO-k	60%	70%
CO4	Design PID controllers and lead-lag compensators using modern computational tools to meet engineering performance specifications.	PO-c, PO-e, PO-k	60%	70%

III. GRADING SYSTEM AND OBE ASSESSMENT MATRIX

Term Weights: Midterm Grade = 50% | Final Term Grade = 50%

Passing Percentage Cutoff: 60% raw score corresponds to a grade of 3.00 (PASSED).

Term	**Assessment Category**	**Weight (%)**	**Mapped COs**	**Assessment Items & Max Score**
Midterm	Midterm Quizzes	30%	CO1, CO2	Quiz 1: Laplace & Transfer Functions (30 pts, CO1); Quiz 2: Block Diagrams & SFG (30 pts, CO1); Quiz 3: Time Response & Stability (40 pts, CO2)
Midterm	Midterm Problem Sets / Seatworks	20%	CO1, CO2	ProbSet 1: Electromechanical Modeling (50 pts, CO1); ProbSet 2: 2nd-Order Transient & Steady-State Error (50 pts, CO2)
Midterm	Midterm Examination	50%	CO1, CO2	Midterm Exam Part I (System Modeling) (50 pts, CO1); Midterm Exam Part II (Time-Domain & Routh) (50 pts, CO2)
Final	Final Term Quizzes	30%	CO3, CO4	Quiz 4: Root Locus Analysis (30 pts, CO3); Quiz 5: Bode & Frequency Response (35 pts, CO3); Quiz 6: PID & Lead-Lag Design (35 pts, CO4)
Final	Control Design Project & Simulation	30%	CO3, CO4	MATLAB/Python Root Locus & Bode Simulation (50 pts, CO3); DC Motor PID Controller Design Project (50 pts, CO4)

Term	Assessment Category	Weight (%)	Mapped COs	Assessment Items & Max Score
Final	Final Examination	40%	CO3, CO4	Final Exam Part I (Root Locus & Frequency Response) (50 pts, CO3); Final Exam Part II (Compensator & PID Design) (50 pts, CO4)